

Multiple Choice

1. **Answer:** A
Options: A) reinforcement learning; B) self-supervised learning; C) unsupervised learning; D) supervised learning
Note: Correct option: A - reinforcement learning
2. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
3. **Answer:** D
Options: A) To save computing time during testing; B) To save space for storing the Decision Tree; C) To make the training set error smaller; D) To avoid overfitting the training set
Note: Correct option: D - To avoid overfitting the training set
4. **Answer:** A
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: A - True, True
5. **Answer:** B
Options: A) $P(A|B)$ decreases; B) $P(B|A)$ decreases; C) $P(B)$ decreases; D) All of above
Note: Correct option: B - $P(B|A)$ decreases

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Transform data to zero mean'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, True'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'It runs reasonably slow for multiple linear regression'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'convolutional networks'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def cummulative_sum(test_list): | return (res)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def even_bit_toggle_number(n) : | return n ^ res`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key